Improving Inventory Management: Predicting Minimum Required Stock and Redistributing Medical Supplies Across Departments

Executive Summary

This project explores improving hospital inventory management by predicting the minimum required stock levels for essential medical consumables such as Gloves, Surgical Masks, and IV Drips. The main goal was to optimize the hospital supply chain by identifying patterns in historical data that could support smarter forecasting and more efficient distribution between departments. I focused specifically on consumables, as they require a different management approach compared to durable medical equipment like X-ray machines or CT scanners.

My intent was to make the hospital supply system more efficient by reducing waste, improving stock alignment with real demand, and ultimately saving money. The idea was to redistribute surplus stock from areas of low usage to departments in need, guided by forecasted demand. Although I didn't implement the full redistribution system, I was able to build a working model for forecasting supply needs and identify overstock patterns, laying the foundation for future optimization work.

The Random Forest model showed strong performance with MAE values of 40.22 for Gloves, 11.44 for Surgical Masks, and 42.21 for IV Drips. Key features like Avg_Usage_Per_Day, Current_Stock, and Restock_Lead_Time were essential for accurate forecasting. These results suggest that forecasting demand based on historical trends is a viable first step in optimizing supply chain processes within hospitals.

Introduction

Efficient inventory management is critical in hospitals to balance having enough supplies without incurring excessive costs from overstock. Overstocking can result in waste, expired stock, and storage issues, while understocking can lead to critical shortages and risks to patient care.

This project aimed to find patterns in historical usage data that could be used to predict the minimum required stock for consumables and to explore redistribution opportunities between hospital departments. By improving prediction accuracy, hospitals can make more informed procurement decisions, reduce unnecessary stock, and allocate resources.

Methods

1. Data Sources

This project uses the **Hospital Supply Chain** dataset by **vanpatangan** from Kaggle, which contains five tables covering inventory, financial, staff, patient, and vendor data.

- Inventory Data: Contains information on stock levels of Gloves, Surgical Masks, and IV Drips across hospital departments, including variables like Current_Stock, Max_Capacity, and Restock_Lead_Time.
- Patient Data: Includes clinical and logistical information such as Admission_Date, Discharge_Date, Primary_Diagnosis, Procedure_Performed, Room_Type, Bed_Days,

Supplies_Used, Equipment_Used, and Staff_Needed. These features help contextualize supply usage patterns across different hospital settings.
- Derived Time Features: Time-based features such as Month, Year, and Days_since_start were derived from the Date column to analyze usage fluctuations over time and observe seasonal trends.

2. Data Cleaning

- Converted date columns into datetime format to enable extraction of time-based features.
- Handled missing values, encoded categorical variables, and standardized numerical features.

3. Analysis Method

- Used Random Forest due to its strength with non-linear patterns, resistance to overfitting, and ability to identify important features.
- Evaluated performance with MAE and MSE. The Random Forest model was applied to the cleaned data grouped by quarter, and the final results showed good accuracy for predicting the minimum required stock of consumables like Gloves, Surgical Masks, and IV Drips.
- Performed exploratory analysis to detect surplus by comparing current stock levels with current minimum required stock values. This helped identify departments holding excess inventory that could be redistributed. I also found clear differences in usage patterns between hospital wards (e.g., ICU vs. General Ward), confirming that department-specific distribution strategies are necessary for efficient supply management.

Results

1. Forecasting Results (Random Forest Model)

The Random Forest model was trained on quarterly data to predict the minimum required stock of Gloves, Surgical Masks, and IV Drips. The model demonstrated solid predictive accuracy, with MAE values of 40.22 for Gloves, 11.44 for Surgical Masks, and 42.21 for IV Drips.

The most important features influencing the predictions were:

- Avg_Usage_Per_Day
- Restock_Lead_Time
- Current_Stock

Figure 1: Actual vs. Predicted minimum required stock by quarter for Gloves

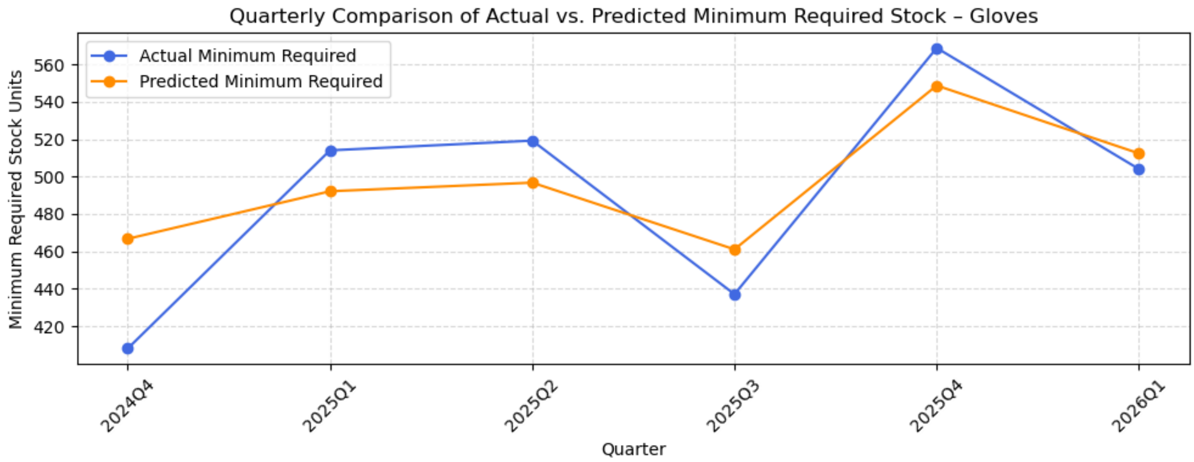


Figure 2: Actual vs. Predicted minimum required stock by quarter for Surgical Masks

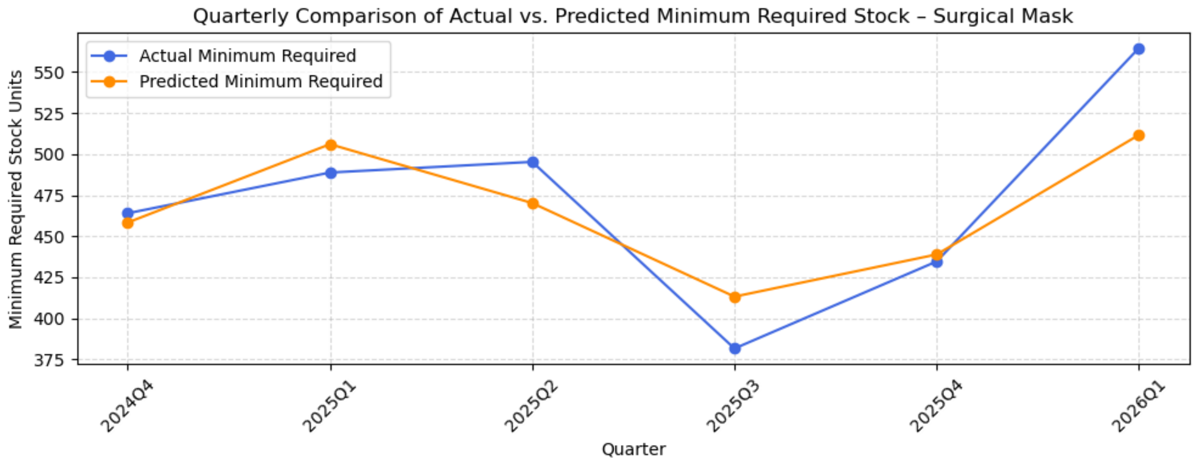


Figure 3: Actual vs. Predicted minimum required stock by quarter for IV Drip

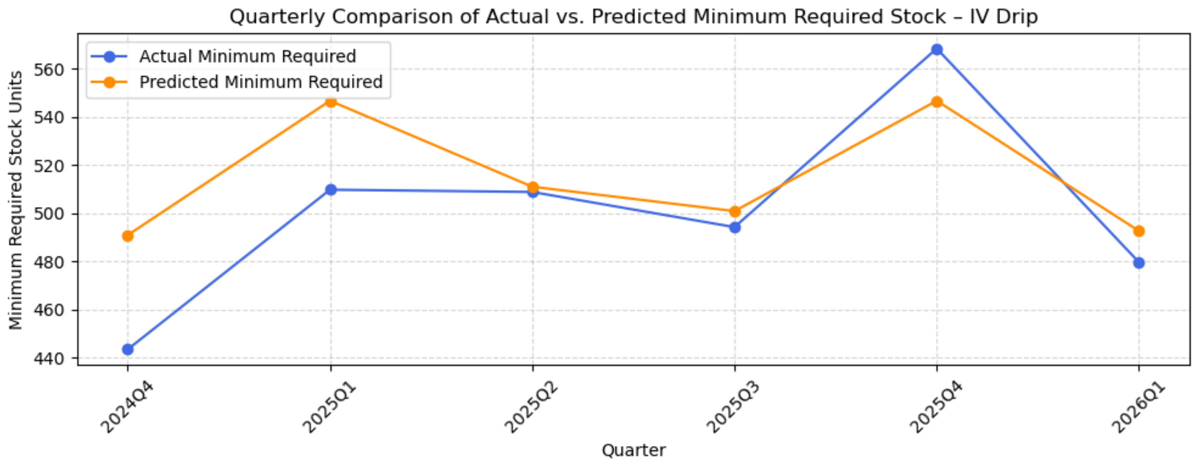
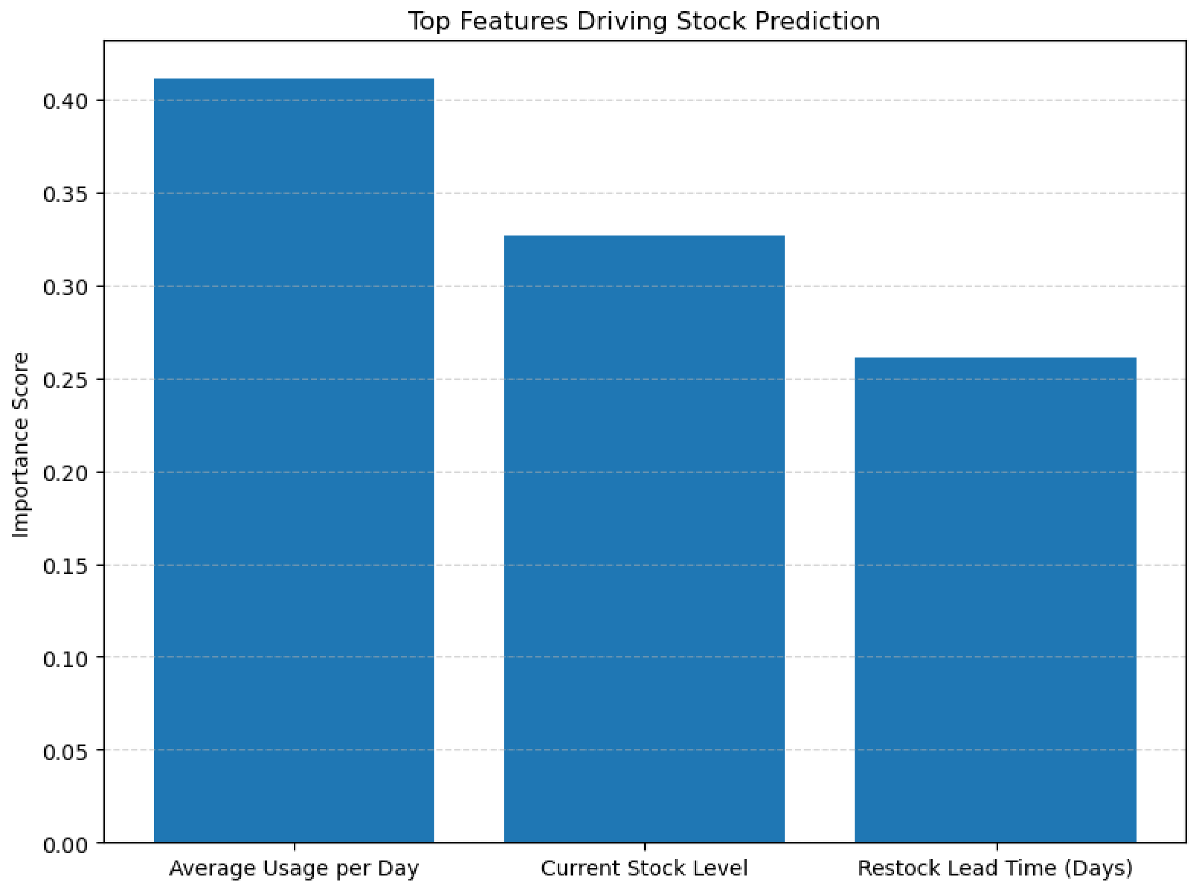


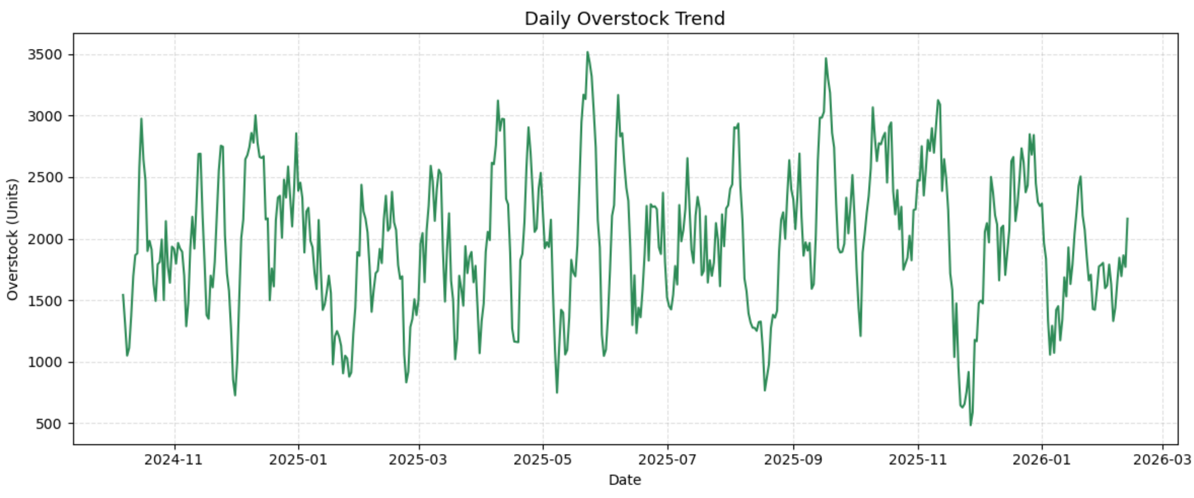
Figure 4: Feature importance from the Random Forest model



2. Overstock Insights

By comparing Current_Stock with the recorded Min_Required values, the analysis revealed a consistent surplus of supplies. This surplus represents an opportunity to reduce waste and optimize inventory.

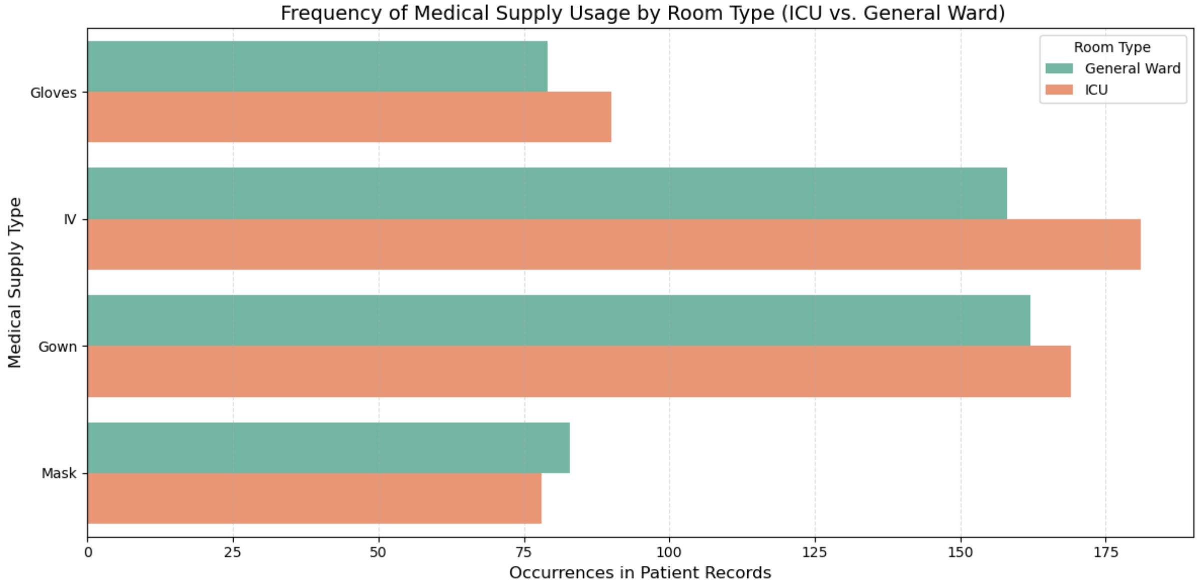
Figure 5: Difference between Current_Stock and Min_Required showing potential overstock of items



3. Department-Level Demand Differences

Further analysis showed that demand for consumables varies significantly by department. For example, ICU and General Ward usage patterns differ depending on the item. This insight is critical for designing an effective redistribution strategy in the future.

Figure 6: Frequency in item usage across departments (e.g., ICU vs. General Ward)



Conclusion

The project successfully predicted the minimum required stock for core hospital supplies and identified areas of overstock that could be optimized. While I didn't reach full implementation of the redistribution strategy, the forecasting model built a strong foundation for future work.

Additionally, although I couldn't complete the redistribution model, the exploratory analysis showed clear potential for using known differences in item demand across wards to guide future redistribution of surplus stock more effectively.

Recommendations for Next Steps

The analysis clearly shows that the hospital has surplus stock, which likely leads to waste. By using forecasting, hospitals can better understand what they really need and plan their supplies more accurately. This can reduce the amount of unused stock and help avoid running out of important items.

Based on this, here are the next steps:

1. **Minimize Surplus and Optimize Distribution**: Use forecasted and current supply insights to reduce excess stock and ensure efficient redistribution based on departmental usage.
2. **Improve Supply Planning with Forecasts**: Make supply decisions based on predicted needs, so future orders match real demand better and resources are used more efficiently.